

# Bulk Canvas → Moodle Migration

A step-by-step guide for administrators

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Migrating many courses off Canvas at once with the **Canvas Uplifter** (`tool_canvasuplifter`) and **Automate** (`tool_automate`) Moodle admin tools.

|                   |                                                                |
|-------------------|----------------------------------------------------------------|
| <b>Applies to</b> | Canvas Uplifter 0.43.0+ and Automate 1.3.0+, on Moodle 5.0–5.2 |
| <b>Audience</b>   | Site administrators and managers running a Canvas migration    |
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# 1. Overview

Canvas Uplifter imports a single Canvas Common Cartridge (.imsc) export into Moodle: it can **analyse** a package (report what it contains and how cleanly each part maps to Moodle) or **build** a brand-new Moodle course from it. For a handful of courses you can drive it by hand from its own upload page.

When you are moving **many** courses, the Automate plugin drives Canvas Uplifter in bulk. You feed it a list of Canvas backup URLs and/or a server directory of packages; Automate turns each package into one background Canvas Uplifter job. This guide covers that bulk workflow end to end.

## The two ways each package can be handled

|                          |                                                                                                                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Build courses now</b> | A new, hidden Moodle course is created from each package automatically. Best when you trust the packages and want courses ready to review.                                             |
| <b>Analyse for later</b> | Each package is fetched and a conversion report produced, but no course is built. An administrator reviews each report and builds the course manually from Canvas Uplifter when ready. |

### Why background jobs?

Nothing is converted inside your browser request. Every package becomes a background adhoc task, so a directory or URL list of hundreds of courses is queued instantly and then worked through by cron instead of timing out the page. They run under Moodle's ordinary site-wide task scheduling — how many run at once follows your *Server > Tasks* worker configuration.

# 2. Before you start

- **Both plugins installed.** Canvas Uplifter provides the import engine; Automate provides the bulk driver. Automate detects Canvas Uplifter automatically — the bulk-import feature only appears when it is present.
- **Capabilities.** You need `tool/automate:manage` to run Automate, and `tool/canvasuplifter:use` for the import itself. Building courses additionally needs `moodle/course:create` in the target category.
- **Cron running.** The imports run as background tasks, so Moodle cron must be active (*Site administration > Server > Tasks > Task processing*).
- **Trusted packages only.** Imported HTML is stored and displayed as authored. Only import .imsc packages from sources you trust.

### Security: server-side URL fetch (SSRF)

When you import from a download URL, the site fetches it server-side through Moodle's SSRF-aware cURL layer, governed by *Site administration > Security > HTTP security*. On Moodle 5.0+ these settings block loopback and private ranges by default. Keep those defaults; uploading the file directly avoids the fetch entirely.

### 3. Turn on bulk Canvas import

The feature ships behind an off-by-default kill-switch. Enable it once:

#### Step 1 — Open Automate's settings

Go to *Site administration > Plugins > Admin tools > Automate > Settings*.

#### Step 2 — Allow bulk Canvas import

Tick **Allow bulk Canvas import**. The background import task re-checks this switch before it runs, so clearing it later also stops any still-pending bulk import from starting — but it does not cancel a Canvas Uplifter job that has already been created, so a build already under way still finishes.

#### Step 3 — (Optional) set a source directory

If you will import from a folder of packages on the server, set **Bulk Canvas import source directory** to that folder (a staging area holding `.imsc` / `.zip` files). You can skip this if you only import from URLs.

### 4. Prepare your packages

#### Option A — A list of Canvas backup URLs

In Canvas, export each course as a Common Cartridge and collect the download URLs — one per line. Each is fetched in the background.

```
https://school.instructure.com/exports/course-101.imsc
https://school.instructure.com/exports/course-102.imsc
https://school.instructure.com/exports/course-103.imsc
```

#### Option B — A directory of packages

Copy the exported `.imsc` files into the source directory you configured in step 3. The picker searches server-side and type-to-filters, so it scales to hundreds of files.

You can use either source, or both together in one run.

## 5. Run a bulk import

### Step 1 — Open the bulk import page

Go to *Plugins > Admin tools > Automate > Bulk Canvas import*.

### Step 2 — Choose your sources

Paste your URL list into the text area, and/or pick packages from the server directory. Preview the selection before queueing.

### Step 3 — Choose what happens to each package

- **Build courses now** — a new course is created from each package (needs course-create permission in the target category).
- **Analyse for later** — each package is analysed only, ready for you to review and build manually.

### Step 4 — Set the target category and options

Pick the course category new courses land in. Optional build settings (such as building a runnable quiz from each standalone question bank, or combining consecutive pages into a book/lesson) carry through to each job.

### Step 5 — Queue the jobs

Confirm to queue every package as a background job. Watch progress in *Site administration > Server > Tasks > Task logs* — the tasks appear as **Analyse Canvas package** and **Build course from Canvas package** (queued under **Bulk Canvas import**).

## 6. Review & build staged imports

Packages you queued with **Analyse for later** appear under *Automate > Staged Canvas imports* once analysed. From there you can:

- **Review a report and build the course** — open a completed analyse job to reach Canvas Uplifter's status page, read the conversion report, and use **Build this course** when you are happy.
- **Jump to a built course** — a completed build links straight to the course it created.
- **See storage used** — the top of the page shows how much space your stored packages occupy.
- **Reclaim space** — tick finished imports and **Delete selected** to free their stored packages.

#### Deleting is safe

Deleting a finished import frees its stored .imsc package **once no other job still references it** (an analyse job and the build made from it share one package, so it is freed only when the last of them is deleted) **but keeps any course a build already created** — it reclaims cache, it does not undo an import. Only finished (done or failed) imports can be deleted, so a delete never interrupts an in-flight conversion.

## 7. Driving it from the command line

The same bulk job can run from the CLI — handy for very large migrations or scripting. Run with `--help` for all options, or `--list` to inspect the sources without queueing.

```
# Build a course from every URL in a file, into category 5:
php admin/tool/automate/cli/import_canvas.php \
--urls=/data/urls.txt --category=5 --mode=build --execute

# Analyse a directory of packages for later review:
php admin/tool/automate/cli/import_canvas.php \
--source=/data/canvas --category=5 --mode=analyse --execute

# Inspect what would be queued, without queueing anything:
php admin/tool/automate/cli/import_canvas.php --urls=/data/urls.txt --list
```

## 8. Monitor & troubleshoot

|                              |                                                                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Where to watch</b>        | <i>Site administration &gt; Server &gt; Tasks &gt; Task logs</i> . Each import narrates what it did.                                                                                |
| <b>A job failed</b>          | Open the job's status page from <i>Staged Canvas imports</i> for the error; failed jobs can be deleted and re-queued.                                                               |
| <b>Quiz has no questions</b> | Canvas sometimes exports a quiz shell without its item bank. It builds as a hidden placeholder; re-export from Canvas with item banks included.                                     |
| <b>A URL will not fetch</b>  | Check <i>Security &gt; HTTP security</i> — the host or port may be blocked. Upload the file directly to bypass the server-side fetch.                                               |
| <b>Feature missing</b>       | Confirm both plugins are installed and <b>Allow bulk Canvas import</b> is ticked. Bulk import needs Canvas Uplifter 0.40.0+; the storage counter and selective delete need 0.43.0+. |

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